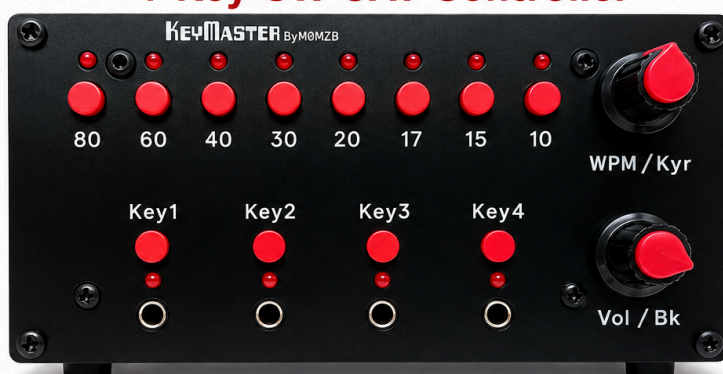


KEYMASTER By MØMZB

4-Key CW CAT Controller



Use paddle, bug, straight key and cootie together



Automatically detects the active key



Remembers radio settings for each key



Instantly sets WPM, keyer, sidetone, band & more via CAT



CAT control for Yaesu, Icom & more



Built for CW operators by MØMZB

TURN YOUR RADIO INTO A CW MASTER STATION

USER MANUAL

Version 20260719

VERSION 1.0

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WARNINGS/ADVICE

Thank you for choosing KeyMaster. I hope it becomes a useful addition to your station. Please read the following information before connecting the unit.

- **DO NOT** Hot swap the serial connectors. The KeyMaster must be powered down (remove the DC jack) before swapping any serial connector. You have been warned, you may get away with it once, twice or even 10 times, but hot swapping serial connectors has the potential to damage the KeyMaster.
- **DO NOT** exceed the recommended voltage (13.8V)
- Although a 5V supply connector is provided on the rear panel, this is primarily as a convenience when uploading firmware. The unit has not been tested for long term operation on a direct 5V supply, but there is no reason it will not work.
- **DO NOT** Connect more than one serial interface at any one time – connect to only one radio at a time.
- **DO** Use the same power supply for the KeyMaster as your radio. If you intend to use a different supply, be careful that you do not introduce a different GND - you could cause significant damage by doing this.

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Introduction

KeyMaster is a smart controller for sharing one radio between four CW key inputs. KeyMaster allows up to four different CW keys to remain permanently connected to a single radio. Each key remembers its own CW speed, keyer mode and break-in setting, allowing instant changes between keys without repeatedly adjusting the radio.

KeyMaster has a unique autosense mode, where the KeyMaster senses which key is in use, and automatically switches to route the active key to the radio. The keys can also be switched manually, using buttons on the front panel.

Each of the **four key inputs on KeyMaster has its own saved operating bank of settings**, so changing from one key to another will also recall the chosen CW speed, keyer mode, and break-in mode for that key. Sidetone/monitor volume and band choice is shared across all banks and remains unchanged when switching keys.

The unit can run standalone with no radio connected. When a supported radio is connected by CAT, KeyMaster sends the front-panel changes to the radio. The KeyMaster polls the radio so any changes made directly on the radio interface are read-back by KeyMaster, and the relevant settings updated.

When adjusting WPM, sidetone volume, Keyer ON/OFF and Break-in ON/OFF the KeyMaster displays the current value using the front panel LEDs as a bar chart, or animation.

The KeyMaster features Band select buttons, allowing the operator to quickly switch between bands on the connected radio. This is especially useful for radio such as the Yaesu FTDX10, where there is no direct access to band selection on the front panel.

The KeyMaster **uses an analogue switch, so there is no latency introduced to CW key use**

With Key Master auto-keying you can

- Have several keys connected, and a different WPM set on each, allowing super quick changes of speed.
- Have a Bug, Paddle, Straight key and Cootie attached all at the same time, and rapidly switch between them
- Have a bug and paddle connected at the same time, use the paddle to sound a phrase, and then try to sound with the bug. This is a great way to practice bug sending, and get your bug fist as close to perfect as possible.
- Directly adjust WPM, Sidetone volume, Keyer, Break-in, and choose a new band – it's like having a top-of-the line radio where all the settings are available on the front panel!

Quick Start Guide

1. Connect the KeyMaster to your radio via one of the serial connectors. For example, direct connection via a standard Serial DB9 connector to an FTDX10.
2. Connect the Key output on the rear panel of the KeyMaster to your radio Key input
3. Connect up to four keys to the KeyMaster
4. Connect to the same power supply as your radio (assuming it's in the range 8 to 13.8V) via the DC barrel jack socket.
5. The KeyMaster will start up, and default to the ftDX10 settings. You can select a different radio by (i) Short press Key 3 to activate Key 3 (even if there is no key connected, this is needed to enter the radio setting mode). (ii) Long press Key 3 for about 3 seconds (iii) Choose your radio with the band switches (see manual for which switch corresponds to which radio. Long press Key 3 again to exit band settings
6. Toggle the LED bar graph mode ON/OFF by selecting Key 2, and then long press on Key 2.
7. Toggle auto key switching by selecting Key 1 and then long press on Key 1
8. Now you are up and running, enjoy!

Power

8–13.8V DC

Connections

- KEY OUT -> Radio Key input
- Radio control (12V and 5V Serial)
- Up to four keys

Short Press

Band buttons Select band

Key buttons Select key

Encoder 1

Rotate =increase/decrease WPM

Press = toggle keyer

Encoder 2

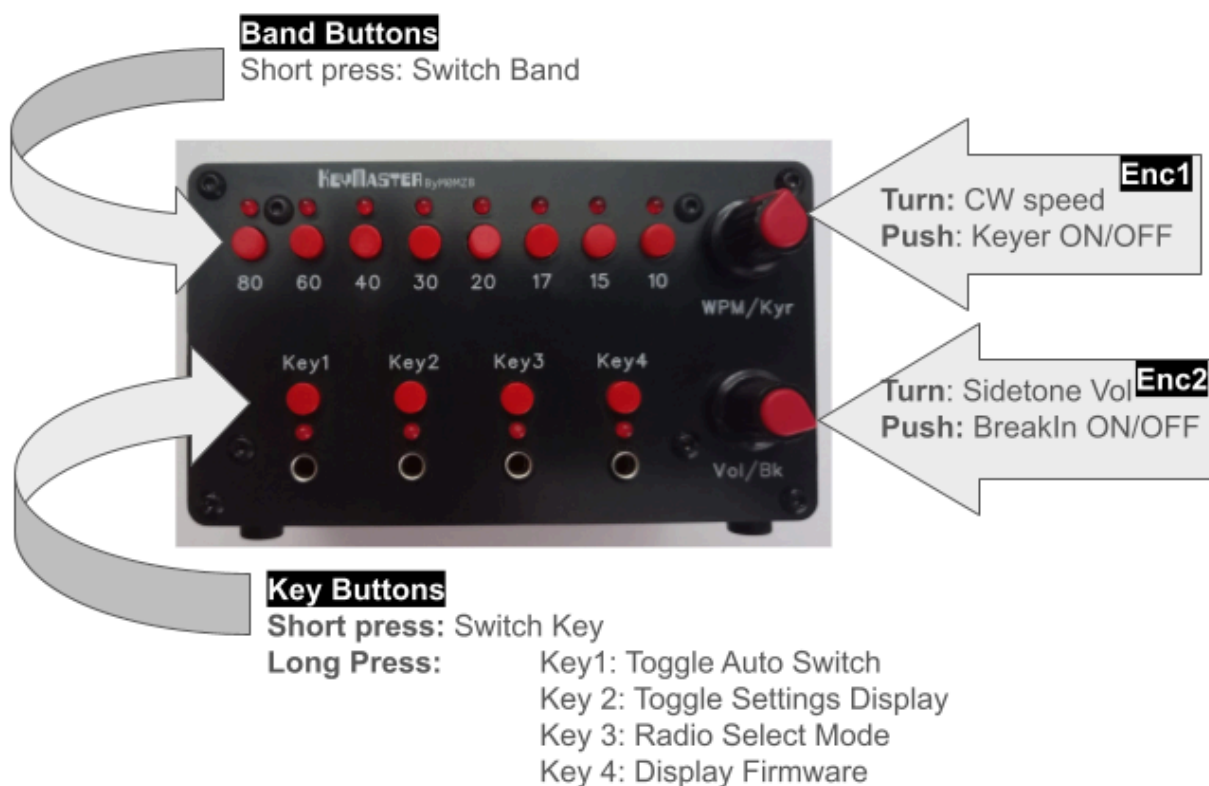
Rotate = Change Sidetone Volume

Press = toggle break-In

Long Press

- Key1 Toggle Auto Switching (auto sensing and switching of keys)
- Key2 Toggle LED Display (animated LEDs and bar graph for WPM, volume, etc)
- Key3 Radio Select Mode (choose radio and baud rate)
- Key4 Display Firmware Version (LEDs display firmware number in binary notation)

Controls Overview



- The KeyMaster includes two rows of buttons.LEDs and two encoders. The top row of buttons allow selection of bands. The LEDs show the chosen band. The Band LEDs are also used as a momentary bar-chart display of settings values, as explained below.
- The Key LEDs/Buttons allow manual selection of the Key to use (manual selection is always available, even if autokey selection is enabled).
- A pair of push-button encoders allow selection of WPM, sidetone volume, Keyer (ON/OFF) and break-in (ON/OFF)
- Four key inputs are provided via 3.5mm sockets. The selected key is routed to the socket on the rear of the unit. An analogue switch is used, so there is no latency introduced.

Band Buttons and Band LEDs

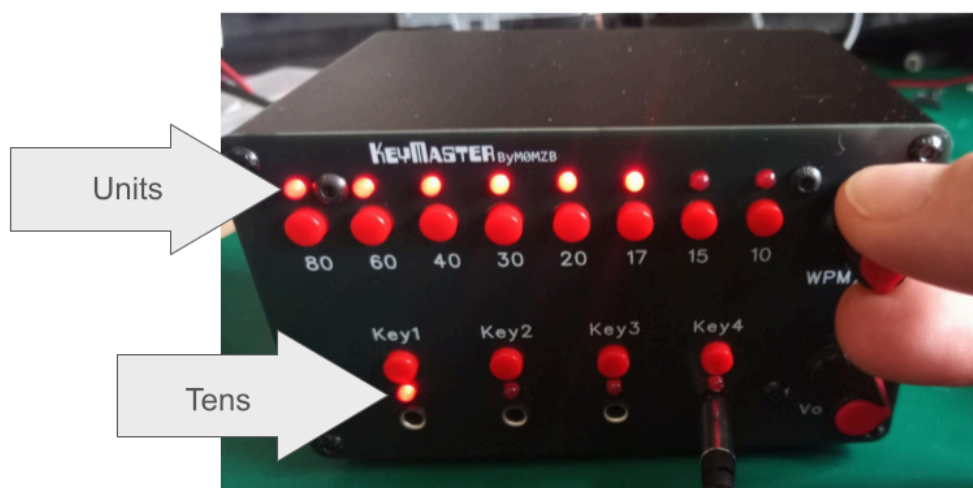
The eight band buttons select:

80m 60m 40m 30m 20m 17m 15m 10m

The small LEDs above the band row show the selected band. Some temporary displays reuse the band LEDs for short animations, then return to the selected band display. Where a band-stack register is available on a radio, the KeyMaster will make use of it (i.e. the radio recalls the previous mode and frequency used on that band).

If the settings display is toggled ON (via Key 3 button, see below) the band LEDs show a momentary bar chart of the WPM or volume value when rotating the WPM or Volume encoder. The Band LEDs show a bar graph from left to right, with the value of "9" indicated by alternate LEDs illuminated.

The tens values are shown by the Key LEDs (Key1 LED =10, Key 2=20 etc.).



Display shows CW speed of 16 WPM

Key Buttons and Key LEDs

The key buttons can be used to manually select the key in operation. The active key is shown by the corresponding LED. When break-in is off, the active key LED flashes with a short off period. When break-in is on, the active key LED stays solid.

The buttons also have a secondary function that can be accessed by holding for three seconds, but only when that key is chosen. The active key LED blinks after about one second of holding as a warning that the three-second action is about to trigger. To access the secondary function, first short press the key button (if that key is not already chosen) and then long press.

Action	Result
Short press <code>Key1</code> to <code>Key4</code>	Select that key/input bank
Hold the currently selected <code>Key1</code> for 3 seconds	Toggle automatic key-based input switching
Hold the currently selected <code>Key2</code> for 3 seconds	Toggle encoder-value display on the LEDs
Hold the currently selected <code>Key3</code> for 3 seconds	Enter or leave radio-selection mode
Hold the currently selected <code>Key4</code> for 3 seconds	Display firmware version, then send callsign <code>M0MZB</code> on the LEDs in Morse

Auto Key Switching

This where the KeyMaster can make your collection of CW keys much more fun! You can have four completely different keys connected to your radio at the same time, and seamlessly switch between them.

With Auto Key Switching Enabled (see above section for instructions on how to do this), the KeyMaster detects which key is in use, and activates that key. A quick press of the left hand paddle (for dual paddles) or closed contact on a straight key is all that is needed for the KeyMaster to select you key.

When the KeyMaster detects a different key being used, relative to the one currently selected. It will firstly disengage all keys, then change the setting on the radio to those stored for the newly selected key (eg turning the Keyer ON/OFF, changing the WPM or changing break-in) and then reactivate the key connection to the radio.

With auto-keying you can

- Have several keys connected, and a different WPM set on each, allowing super quick changes of speed.
- Have a Bug, Paddle, Straight key and Cootie attached all at the same time, and rapidly switch between them
- Have a bug and paddle connected at the same time, use the paddle to sound a phrase, and then try to sound with the bug. This is a great way to practice bug sending, and get your bug fist as close to perfect as possible.

WPM and Keyer Control (Encoder 1)

Action	Result
Turn WPM/Kyr	Change CW speed for the active bank
Press WPM/Kyr	Toggle keyer mode for the active bank

When encoder-value display is enabled, changing WPM briefly shows the value on the band LEDs before the display returns to normal.

Keyer feedback on the band LEDs: when selecting Keyer ON/OFF the band LEDs provide visual feedback.



Toggling the Keyer OFF
Display shows a single line
(ie a straight key)



Toggling the Keyer ON
Display shows a pair
Of lines

(ie paddles)

Volume and Breaking Control (Encoder 2)

Action	Result
Turn Vol/Bk	Change the shared monitor/sidetone volume
Press Vol/Bk	Toggle break-in for the active bank

When encoder-value display is enabled, changing volume briefly shows the value on the LEDs before the display returns to normal.

Break-in feedback on the band LEDs: when selecting Keyer ON/OFF the band LEDs provide visual feedback. When break-in is off, the active key LED flashes with a short off period. When break-in is on, the active key LED stays solid

Settings are saved automatically, but not instantly. Leave the unit powered for at least 30 seconds after changing settings if you want to be certain the latest changes have been written.



Toggling the Break-In OF
The Band LEDs show a brief
animation.
The selected Key LED will
blink whenever Break-In is
OFF



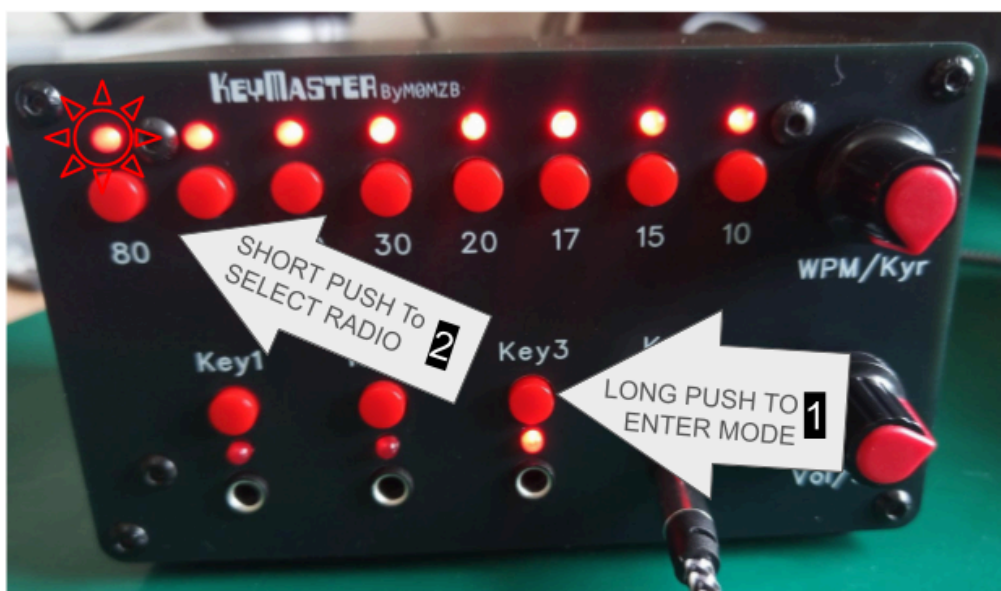
Toggling the Break-In ON
The Band LEDs show the
reverse animation.
The selected Key LED is solid
whenever Break-In is ON

Adjusting Radio Communication Settings

Selecting the radio

Each radio typically requires a unique set of commands, which have been pre-stored in the KeyMaster firmware. KeyMaster can support up to 255 distinct radios. The following steps allow selection of the radio type:

1. Short press `Key3` so bank 3 is the active bank.
2. Hold `Key3` for 3 seconds to enter radio-selection mode.
3. All band LEDs illuminate. The LED for the current radio flashes.
4. Press one of the band buttons below.
5. Hold `Key3` again for 3 seconds to leave radio-selection mode.



Band button	Radio driver
80 / button 1	Yaesu FTDX10
60 / button 2	QRP Labs QMX
40 / button 3	Kenwood TS-590S
30 / button 4	Kenwood TS-590SG
20 / button 5	Yaesu FT-710
17 / button 6	Kenwood TS890S

The radio choice is saved to EEPROM on the next scheduled settings write.

Supported Radios

Radio	Default CAT rate	Implemented status/control
Yaesu FTDX10	38400 baud	Band, CW speed, keyer, monitor volume, break-in
Yaesu FT-710	38400 baud	Band, CW speed, keyer, monitor volume, break-in
QRP Labs QMX	9600 baud	Band and CW speed status; band, CW speed, keyer, monitor volume, and practice/break-in control
Kenwood TS-590S	9600 baud	Band, CW speed, keyer, and monitor status; band, CW speed, keyer, monitor, and break-in control
Kenwood TS-590SG	9600 baud	Band, CW speed, keyer, and monitor status; band, CW speed, keyer, monitor, and break-in control
Kenwood TS-890S	9600 baud	Band, CW speed, keyer, monitor, and break-in status/control

Yaesu FT818, and Icom IC7300 support in future versions. Selecting an unsupported type falls back to the FTDX10 driver.

Setting the CAT Baud rate

The baud rate is set while in radio-selection mode.

1. Short press `Key3` so bank 3 is the active bank.
2. Hold `Key3` for 3 seconds to enter radio-selection mode.
3. All key LEDs illuminate. The LED for the current baud rate flashes.
4. Press one of the key buttons, shown below, to choose the baud rate.
5. Hold `Key3` again for 3 seconds to leave radio-selection mode.

Key button	CAT baud rate
Key1	4800
Key2	9600
Key3	19200
Key4	38400

Encoder Direction Setting

The direction of each encoder can be reversed independently. The usual setting is for clockwise rotation to increase values, but this can be reversed if desired. The LED bar graph is unaffected by reversing the encoder direction.

Encoder	How to reverse its direction
WPM/Kyr	Hold its push-button while powering on
Vol/Bk	Hold its push-button while powering on

Keep the button held throughout the startup LED chase. When the confirmation LEDs appear, release it. The outermost key/band LEDs mean that the selected encoder is now reversed; the innermost LEDs mean normal direction. Repeat the same procedure to toggle it back. Each direction setting is saved immediately and remains in effect after power-off.

Default Settings

On blank or invalid EEPROM, KeyMaster starts on input 1 with automatic key switching enabled, encoder-value display disabled, shared monitor volume 20, and the FTDX10 driver selected.

Bank	CW speed	Keyer	Break-in
1	25	On	Off
2	35	On	Off
3	25	Off	Off
4	35	Off	Off

LED Messages Summary

Display	Meaning
One key LED solid	Active key/input bank
Active key LED flashing	Break-in is off for the active bank
One band LED solid	Active band
All band LEDs on, one flashing	Radio-selection mode
Band LEDs show a temporary bar graph	Encoder-value display for WPM or volume
Every other band LED and every key LED lit briefly	CAT command queue overflow warning
Key LEDs flashing during a button hold	Long-press warning

The overflow warning means the firmware could not queue a CAT command at that moment. The unit continues running; if the radio is disconnected or not responding, KeyMaster should still operate as a standalone key/input controller.

Rear Panel Connections



- A standard 5.5x2.1mm DC jack is used to provide power (user supplied)
- A standard DB9 serial cable is used to connect the 12V Serial to a radio. This connection is perfect for the Yaesu FTDX10 which has a DB9 connector on the rear. Just connect the radio and KeyMaster with a standard DB9 Serial cable
- 5V serial is provided, with a screw terminal plug supplied with the KeyMaster to allow easy make-up of your own cables to suit any radio. 5V Serial is also provided via a 3.5mm socket (Tip=Tx, Ring=Rx and Sleeve=GND). It is very easy to make your own cable with readily available 3.5mm plugs.
- The Key output is a 3.5mm socket that connects to the radio Key input

Troubleshooting

Symptom	Things to check
Radio does not respond to controls	Confirm the selected radio type, CAT cable, radio CAT baud rate, and that the radio is powered
KeyMaster works until the radio is unplugged	This should not happen in current firmware; if it does, note the LED pattern and selected radio type
Active key LED keeps flashing	Break-in is off for that bank; press Vol/Bk to toggle it
The wrong key bank is selected when keying	Check whether automatic key-based input switching is enabled
A long press does nothing	Make sure you are holding the currently selected key button, not an inactive bank button
Latest setting was lost after power-off	Leave the unit powered for 30 seconds after changing settings
Uploading firmware fails	Disconnect or isolate the radio CAT interface while programming